Introduction presentation Festo in the Process industry

A short presentation about Festo’s portfolio



Festo - A leading world wide supplier of industrial automation

Festo founded in **1925**

Festo subsidiaries in **62** countries

› 1000
Sales engineers worldwide

R&D quota
8% of total revenue

100 New patents annually

Catalogue components
33 000

24 h
Delivery service

40
Years of Festo Didactic

Turnover **3.2 Bn €**

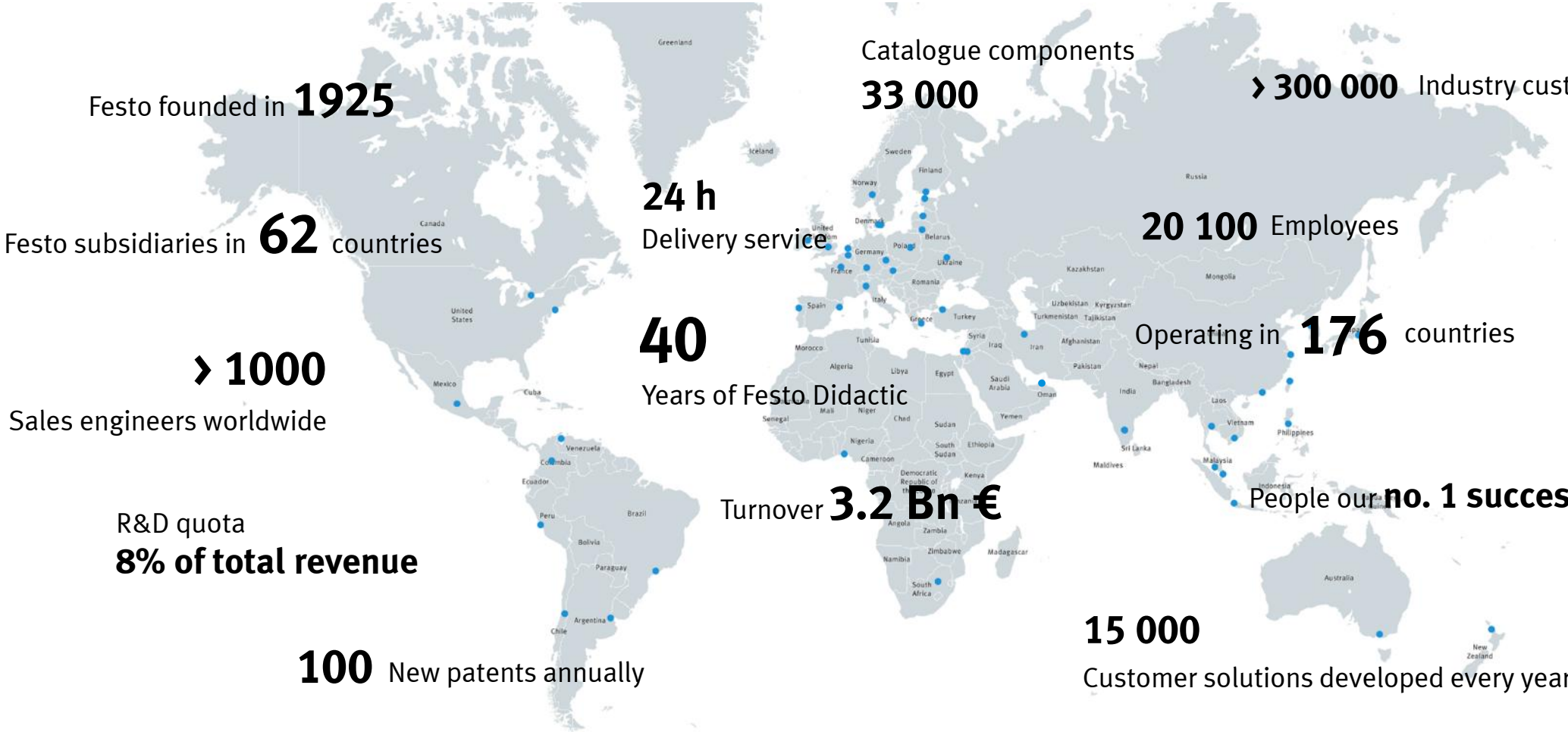
› 300 000 Industry customers

20 100 Employees

Operating in **176** countries

People our **no. 1 success factor**

15 000
Customer solutions developed every year

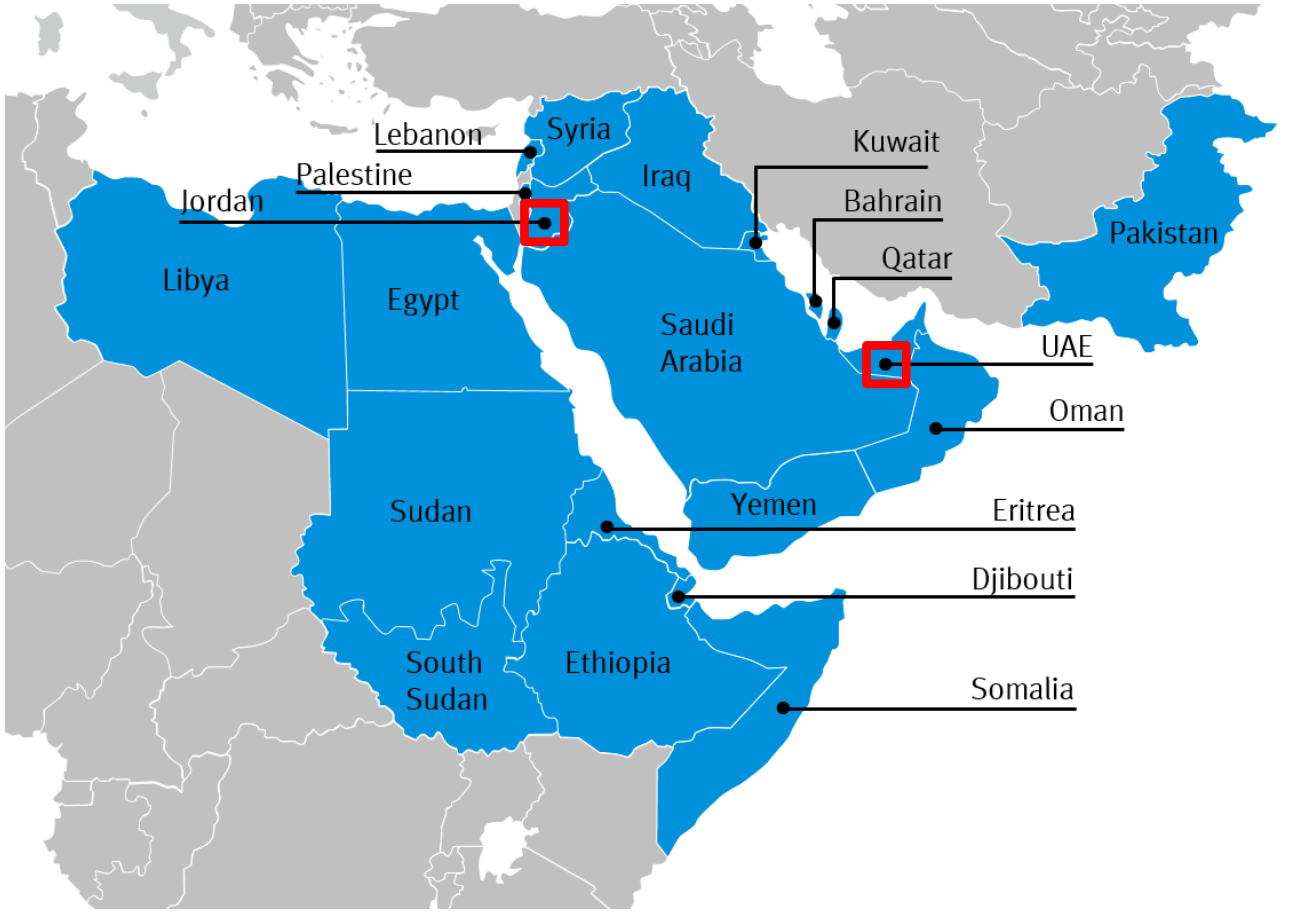


Festo presence in MENA

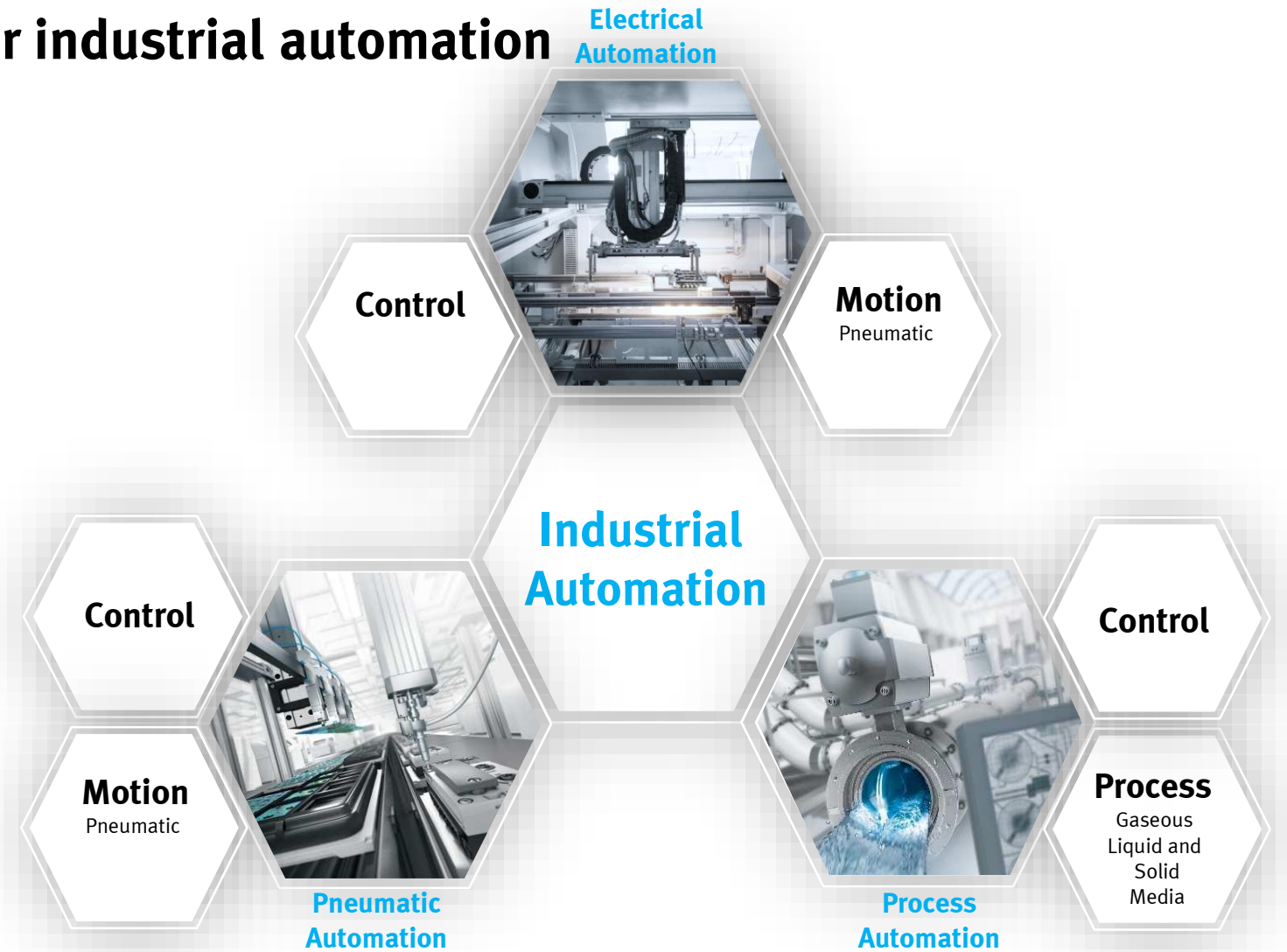
1. Festo Regional Offices in Dubai and Amman
- Regional Managing, technical support and development.

2. Strong and stable network of agencies and dealers
- Since more than 40 years presence
- With more than 15 branches
- More than 100 engineers trained for Festo products & service

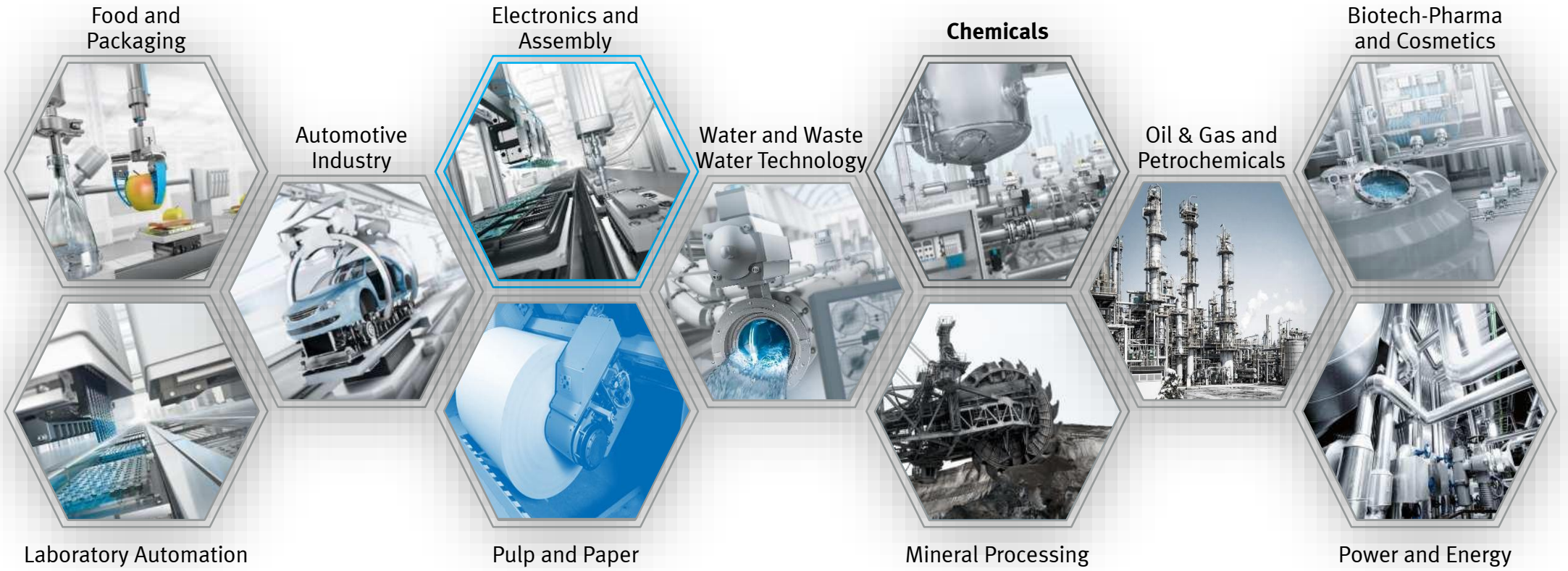
3. Festo HQ Germany
- Manufacturing
- Customized designs
- Central Warehouse
- Other support functions



We are experts for industrial automation



Focus Industry Segments



Factory Automation

Process Industries

Where do we focus and what do we offer?

Where are we active?



Food & Beverage



Water Industry



Chemical Industry



Pulp & Paper



Pharma & Cosmetics



Mineral Processing

What do we control?

Liquids
e.g. Beverages
Oil
Water
Paint
Olefins, etc.

Solids
e.g. Paper
powders
Granulates
Ore
Sludges, etc.

Gases
e.g. Steam
Synthesis gas
Exhaust gases
Nitrogen
HVAC, etc.

How do we control?

Process valves:



Ball and Butterfly valves



Individual valves



Knife gate valves

Actuation:



Quarter turn actuators



Linear cylinders

Control and feedback:



Solenoid valves



Limit switches / boxes



Positioners

Sensors and accessories:



Flow & Pressure sensors



Cables, tubing and fittings



Filter regulators

Valve assemblies simplified and Customized Solutions

Ready to install valve assemblies



Features

HART COMMUNICATION PROTOCOL

Ex

IEC IECEx

SIL certified according IEC 61508

Modbus

PI PROFIBUS • PROFINET

NAMUR

ePLAN

NACE MR0175

Engineered solutions

1. Components pre-assembled

Multiple items with single part no.
Ready to install



2. Customized assemblies

High temperature / low temperature
Collection of process parameters
Integrated Modulating control
Specific tubing requirements
High safety demand



3. Pneumatic Control Cabinets

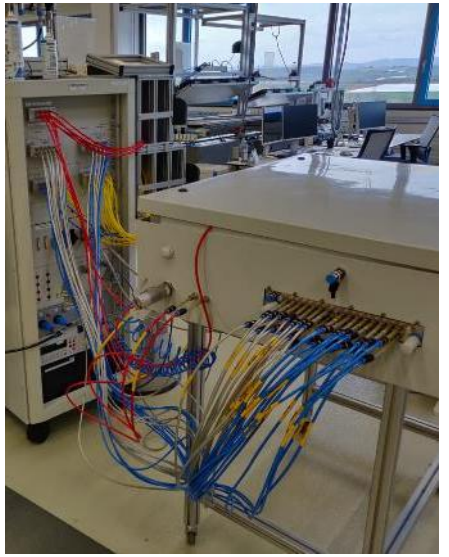
Basic/tailored design to project requirements
Direct or fieldbus communication
Pneumatic valve terminals
Digital / analogue signals
Multiple enclosure styles
Technical documentation



Services and support

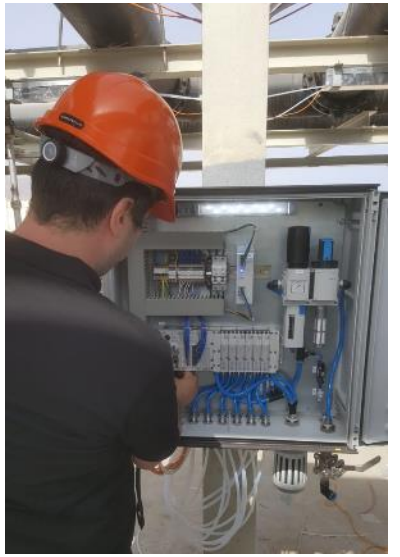
FAT testing and freight





Ensure full functioning equipment prior shipment
- Check as-build design and drawings
- Dynamic function testing
- 3rd party inspectors are welcome
- Freight shipment to your site location*

On site commissioning support





Local support with system start-up
- Reduction of start-up time
- Avoid initial installation mistakes
- Support with integration 3rd party “PLC” systems
- Training sessions of local staff on delivered equipment

* Incoterms to be confirmed per country

Festo's portfolio for process automation – A quick overview

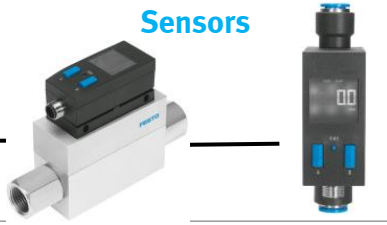
Automation components

Display & Detection

Operator units



Sensors



Controls

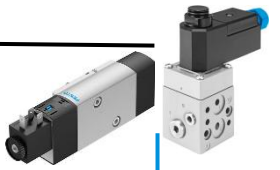
PLC



Valve terminals / RIO



Pilot valves



Valve automation



Positioners



Position transmitters



Linear and quarter turn Actuators

Process valves



Solutions



Process valves



Sub-assemblies



Control cabinets

Accessoiries



Air preparation



Connection technology

Field automation level – Component's

Field level

Solenoid valves


Pilot valve VSNC (NAMUR)


Pilot valve VOFC (NAMUR)


Pilot valve VOFD (NAMUR + thread)


Pilot valve MFH


Pilot valve VUVS


Directional control valve VUVG


Proportional pressure regulator VPPM/VPPX

Sensors and positioners


Sensor box SRBC


Sensor box SRBE (flameproof enclosure)


Sensor box SRBG (inductive double sensor)


Analogue sensor box SRAP


Positioner for quarter turn and linear actuators CMSX


Proximity sensor SME/SMT


Pressure sensor SPAW for gases/liquids


Pressure transmitter SPTW for gases/liquids


Flow sensor SFAB for compressed air/nitrogen


Flow sensor SFAW for fluids

Actuators


Quarter turn actuator DFPD


Quarter turn actuator DAPS


Quarter turn actuator, stainless steel DAPS-CR


Quarter turn actuator with handwheel DAPS-MW


Heavy-duty quarter turn actuator DFPD-HD


Linear actuator DLP


Closed-loop controlled linear actuator DFPI


Closed-loop controlled linear actuator to ISO 15552 DFPI-NB3

Process and media valves


Modular angle seat valve VZXA


Angle seat valve VZXF


Pinch valve VZQA (NO/NC)


Directly actuated valve VZWD


Piloted valve VZWM/VZWP


Force pilot operated valve VZWF


Reverse jet pulse valve VZWE


Ball valve VZBA/VZBM


Ball valve VZBC/VZBD


Butterfly valve VZAF/VZAv


Knife gate valve VZKA

Accessories


Service units


Air dryers


Couplings


Push-in fittings


Plastic tubing


Polytetrafluoroethylene tubing PTFEN


Pressure booster DPA


Compressed air reservoirs

Solutions


Butterfly valve assemblies


Flanged ball valve assemblies


Threaded ball valve assemblies


Knife gate valve assemblies

Operator and RIO level – Component's

Operator level


Front-end display FED (pushbuttons)


Front-end display CDPX 4" (touchscreen)


Front-end display CDPX 7" (touchscreen)


Front-end display CDPX 10" (touchscreen)


Front-end display CDPX 13" (touchscreen)

Control level

Controllers


Integrated controller CPX-CEC


Compact controller CECC


Integrated controller CDPX-CEC

Remote I/O


Modular valve terminal CPX/MPA


Modular electrical terminal CPX


Modular electrical terminal CPX-P


Input/output module CP


Electrical connecting block CAPC


Operator unit CPX-MMI

Valve terminals


Modular valve terminal CPX-P/MPA


Modular valve terminal CPX/VTSA


Valve terminal CPV


Valve terminal CPV Ex-i for use in Ex zones


Valve terminal MPA-S


Valve terminal MPA-L


Valve terminal MPA-C (Clean Design)


Valve terminal VTUG


Valve terminal VTUG optimised for control cabinet installation


Valve terminal VTSA/VTSA-F

Solutions



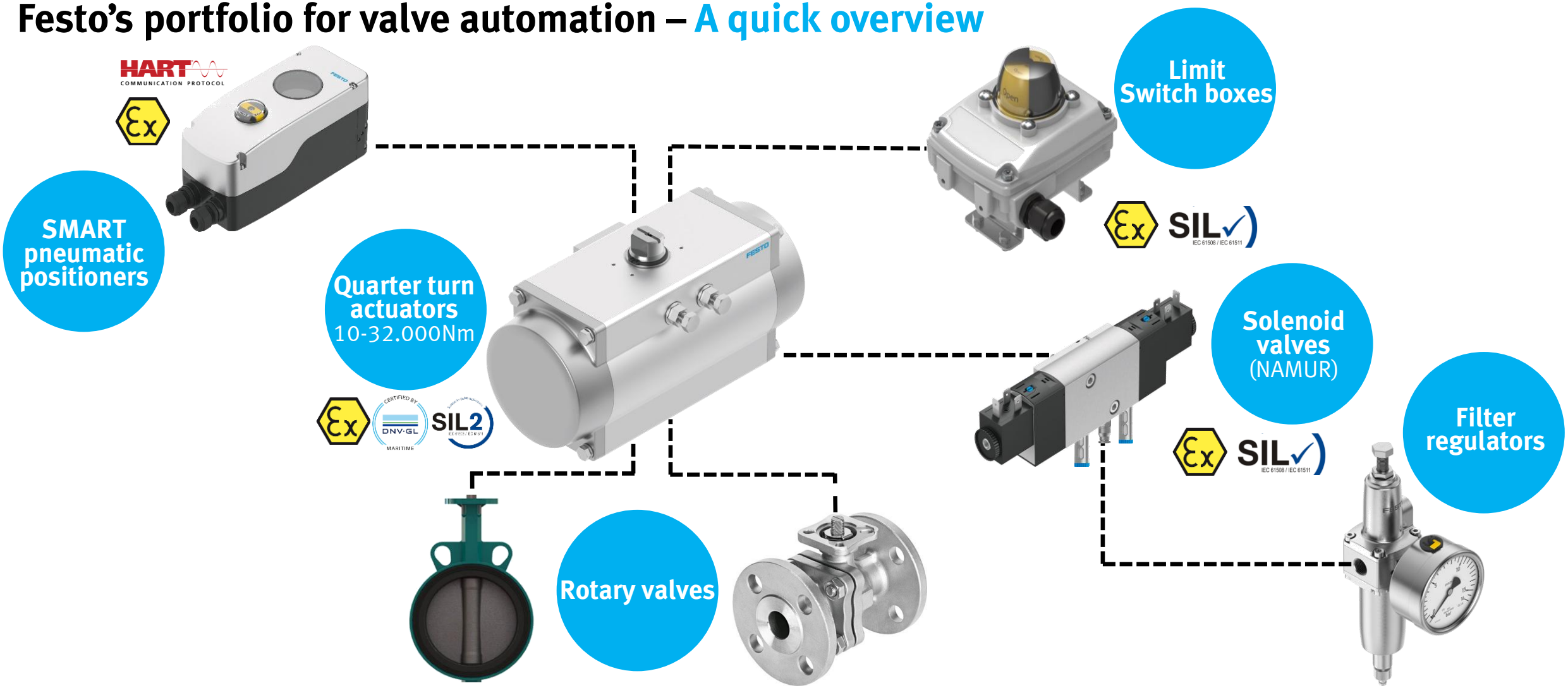


Filter control desks



Pneumatic control cabinets

Festo's portfolio for valve automation – A quick overview



Festo's portfolio for linear valve automation – A quick overview

Pneumatic Positioners (Modulating)



Proximity Sensor



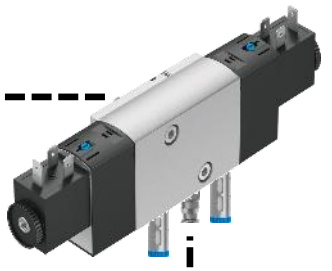
Linear actuators



Displacement encoder



Solenoid Valves (On/Off)



Air reparation



Linear valves





Automation concept – Centralized

Benefits:

- Typically used for small installations with short distances
- Effective for ATEX and SIL applications
- Standardization with NAMUR products needed
- Knowledge level of maintenance staff can be basic (Mech.)

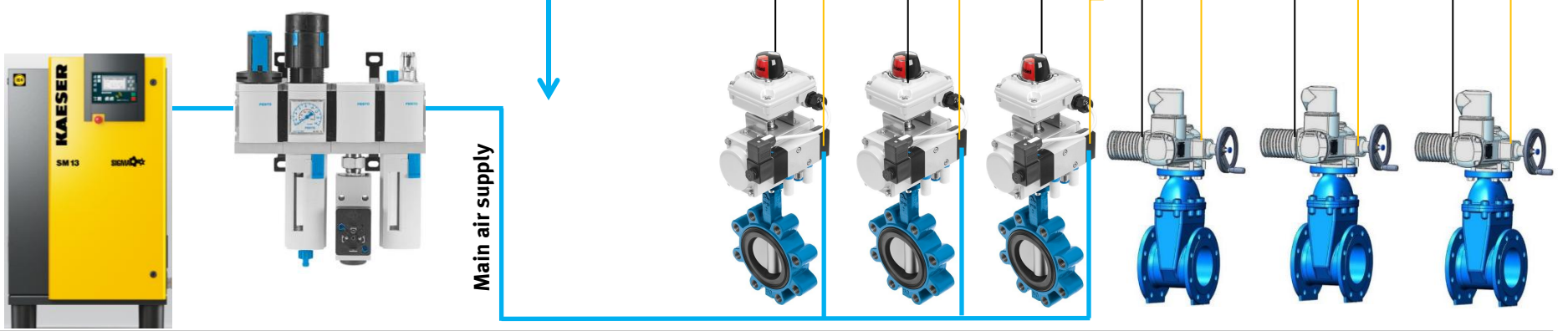
Example:

- For 8 automated valves
- Single coil solenoids
- Limit switch boxes
- Distance control room to Valve assemblies: 100m

Basic estimations:

Digital inputs:	16 DI
Digital outputs:	8 DO
Total cable length:	16*100m
Total tube length:	+/- 50m
Installation time:	high





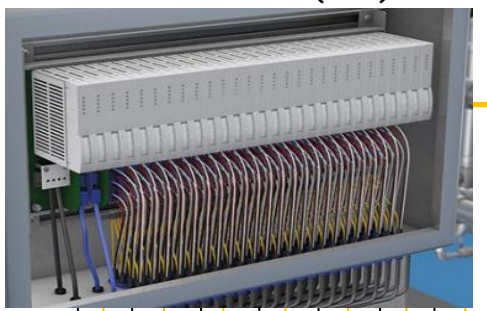
Automation concept – Partially De-centralized

Benefits:

- Reduced cable length and installation time
- Reduce no. of IO cards in main cabinet
- Plant upgrades are made easily
- Extra features like Human machine interface
- All IO functions are shared over the same platform

Basic estimations:

Digital inputs:	16 DI
Digital outputs:	8 DO
Total cable length:	16*10m
Total tube length:	+/- 50m
Fieldbus cable:	1x90m
Installation time:	Medium



90m



Main PLC cabinet

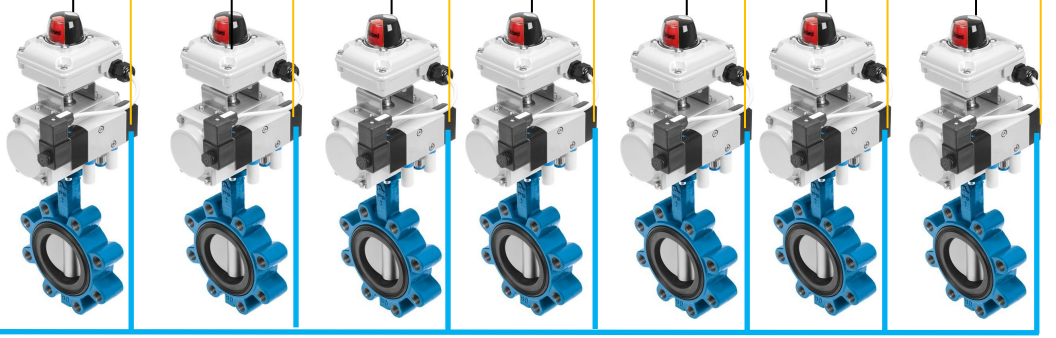
Electric:





100 meters

Main air supply



Example:

- For 8 automated valves
- Single coil solenoids
- Limit switch boxes
- Distance control room to Valve assemblies: 100m

Automation concept – De-centralized

Benefits:

- Reduced cable length and installation time
- Reduce no. of IO cards in main cabinet
- Manual override is centralized
- Plant upgrades are made easily
- Extra features like hot-swap, or Human machine interface
- All IO functions are shared over the same platform
- Trouble shooting with diagnostics can be done remotely
- High level of protection by a metal or GRP enclosure

Example:

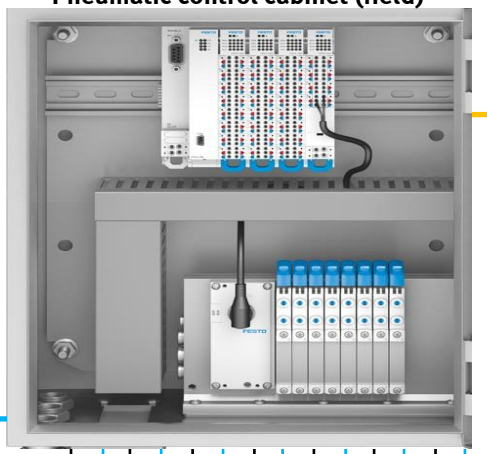
- For 8 automated valves
- Single coil solenoids
- Limit switch boxes
- Distance control room to control cabinet: 90m

Basic estimations:

Digital inputs:	16 DI
Digital outputs:	-- DO
Total cable length:	8x10m
Total tube length:	8x10m
Fieldbus cable:	1x90m
Installation time:	Low







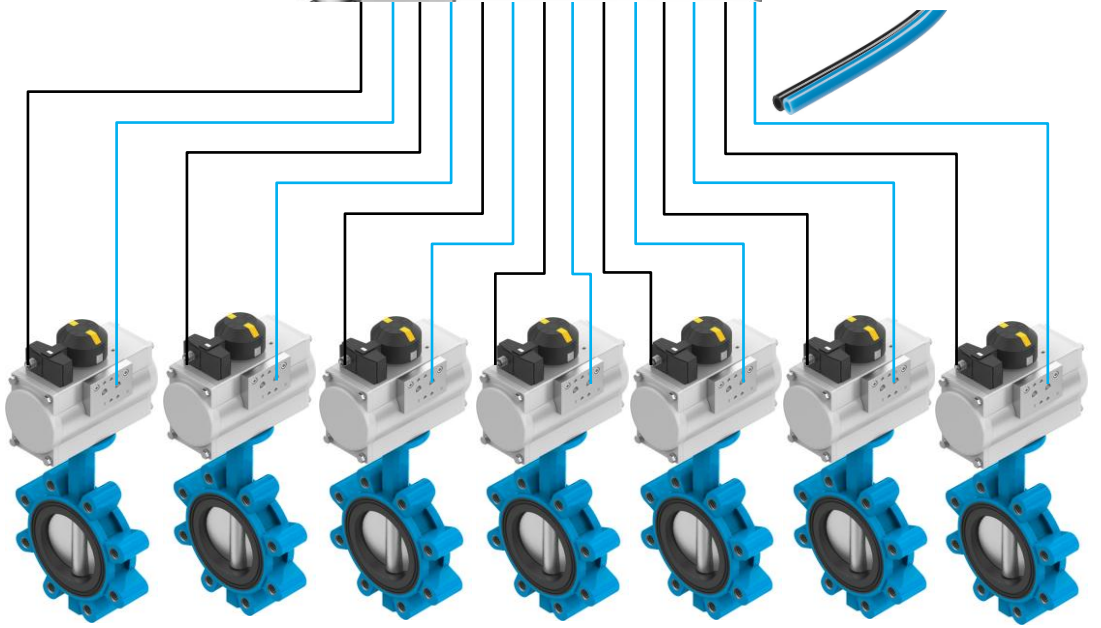
90m



Main PLC cabinet

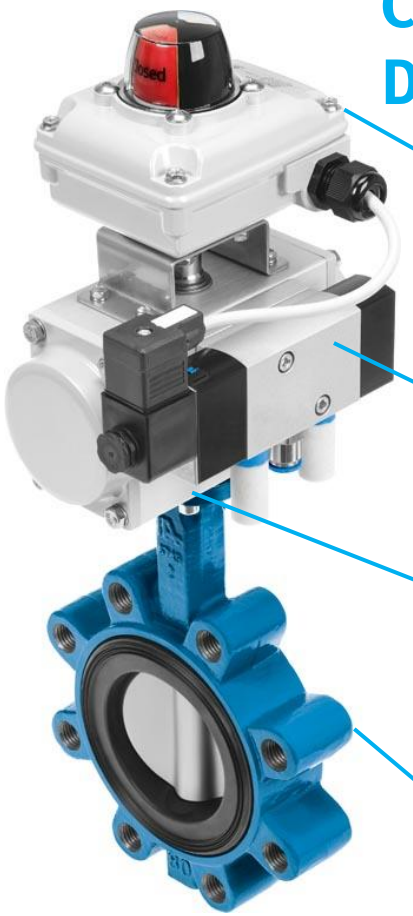
Main air supply

10 meters



Focus topics – Differences for valve assemblies

Centralized / partially De-centralized



Limit switch box
24VDC or 230VAC

Namur solenoid valve
24VDC / 230VAC
Manual override
Speed control fittings

Pneumatic actuator
Double or single acting
Torque up to 32.000Nm

Butterfly valve
Large range of features

De-centralized

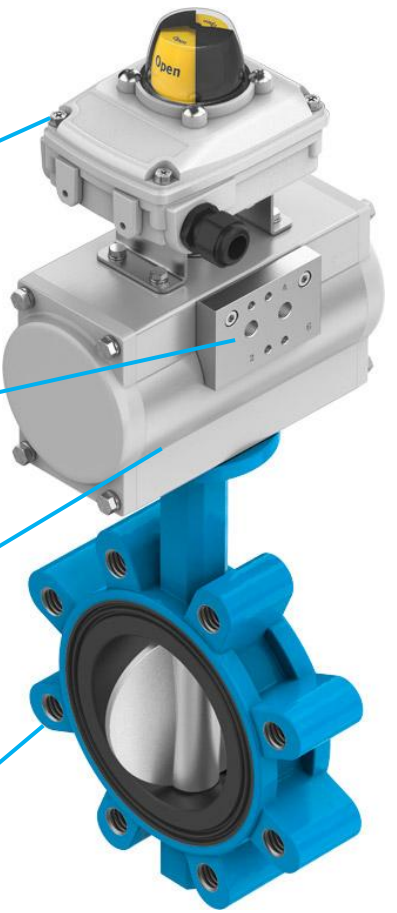
Limit switch box
24VDC or 230VAC

Accessories
Speed control fittings



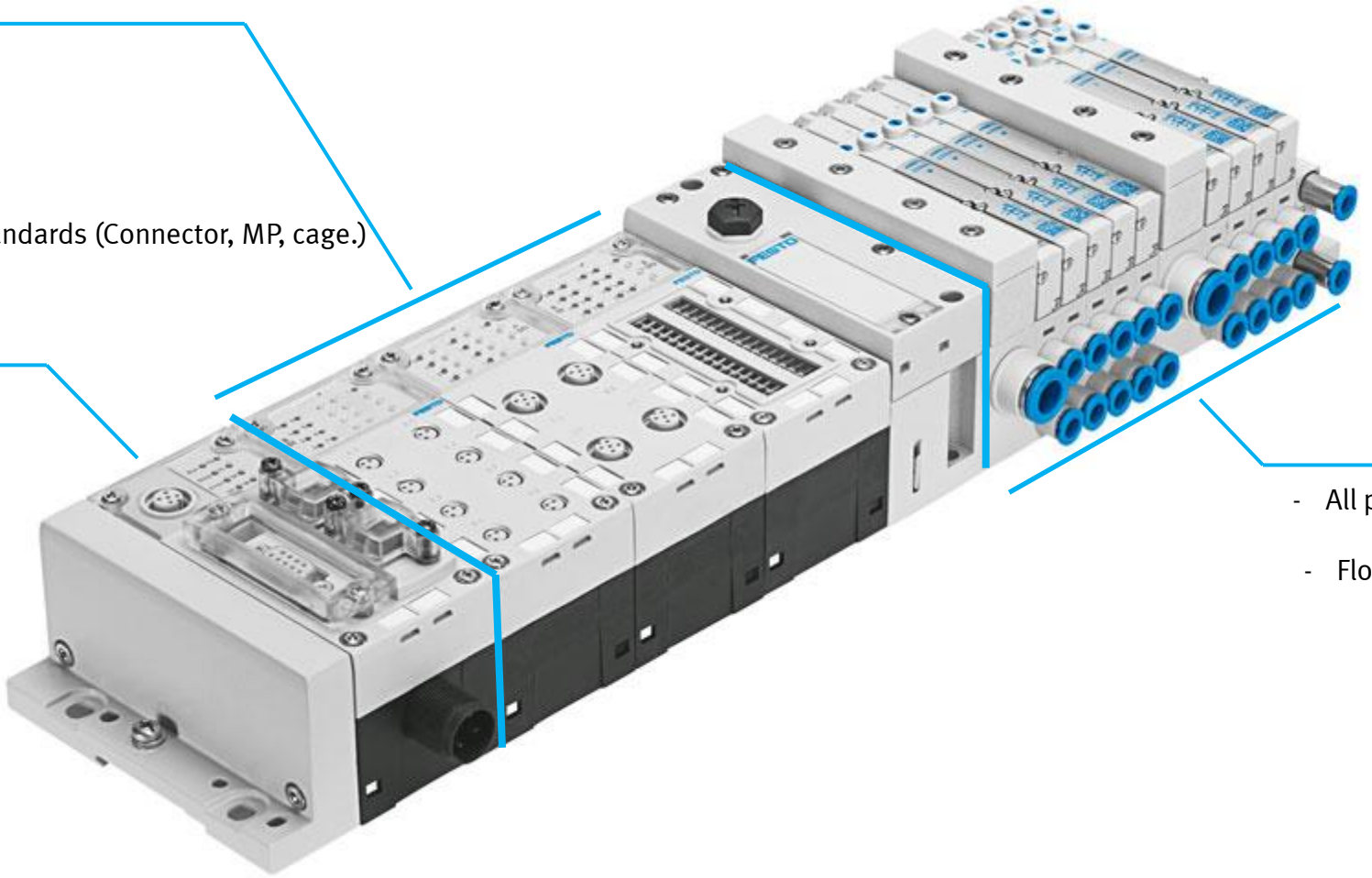
Pneumatic actuator
Double or single acting
Torque up to 32.000Nm

Butterfly valve
Large range of features



Automation concept – Valve terminal technology CPX-MPA

Electrical interface:
- Digital inputs
- Digital outputs
- Analogue inputs
- Analogue outputs
- Pressure reading
- Multiple connection standards (Connector, MP, cage.)



Fieldbus node:
- Connection with main PLC
- Major industry communication protocols available
- Optional features with PLC function based on CodeSyS
- System diagnostics

Pneumatics:
- All pneumatic valve functions: 5/2, 3/2, 5/3
- Flow range up to: 4.000l/min
- Modulair build up
- Vacuum switching
- Proportional air control
- Hotswap available

Summary– Festo a turnkey system provider

Alignment of all system requirements:



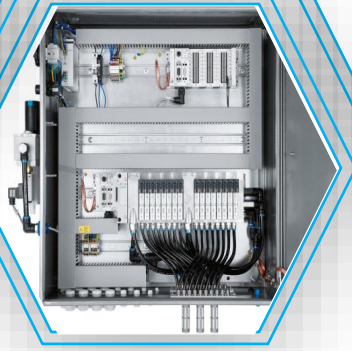
Air generation



Air quality
ISO8573-1:2010



Piping design



Control/cabinet



Tube & fitting



Valve actuation



Installation and
services (ISO-11011)

Thank you for your time

How can we follow up on actions for a possible collaboration?

